

Pure Music

Vocal Isolation for Music

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Abstract. This report presents a study on deep learning models for vocal isolation in music, employing a Dual-Path network architecture inspired by SpeechBrain. We conducted three experiments using the MUSDB18 dataset and a custom-prepared subset. The first experiment established a baseline with short excerpts, confirming pipeline functionality but revealing limited separation quality due to data and computational constraints. The second experiment investigated the impact of model dimensionality and training duration on longer audio samples, finding that extended training improved performance even for lower-capacity models, and that increased dimensionality alone was insufficient. The third experiment explored the effect of separation complexity by comparing two-stem (vocals, drums) versus four-stem (vocals, drums, bass, other) separation. Results indicated a trade-off, with the four-stem model showing potentially better overall source preservation with longer training, but possibly increased interference compared to a model focused on fewer sources. The study highlights the crucial roles of dataset size, computational resources, model architecture, training duration, and the complexity of the separation task in achieving effective vocal isolation. Future work should address these limitations to advance the field.

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1 Goal of the Project

The primary goal of this project was to design, implement, and evaluate a deep learning model for the task of vocal isolation in music, effectively separating vocal tracks from their instrumental accompaniment. This involved leveraging the MUSDB18 dataset and adapting existing source separation techniques, drawing inspiration from the SpeechBrain toolkit. A key objective was to investigate the impact of the separation complexity by comparing the model's performance on isolating a limited number of sources versus all available stems. Additionally, the project aimed to explore the influence of different model architectures and training durations on the quality of vocal isolation.

1.1 Experiment 1: Baseline Vocal Isolation Model

1.1.1 Motivation

The goal of this initial experiment was to establish a fully operational training pipeline for vocal isolation using a small, readily available subset of the MUSDB18 dataset. The musdb package provides 7-second excerpts from the full dataset, which are automatically downloaded and ideal for rapid prototyping and verifying the functionality of the training script and data loading mechanisms. This baseline experiment aimed to ensure all components of the training process, including data ingestion, model forward and backward passes, loss calculation, and optimization, were correctly implemented and functioning on a manageable scale before moving to larger datasets and more complex experiments.

1.1.2 Methodology

The baseline model setup, inspired by SpeechBrain's LibriMix recipe, employed a Dual-Path network architecture. This model was adapted to the available computational resources while maintaining consistency for subsequent experiments to ensure fair comparison.

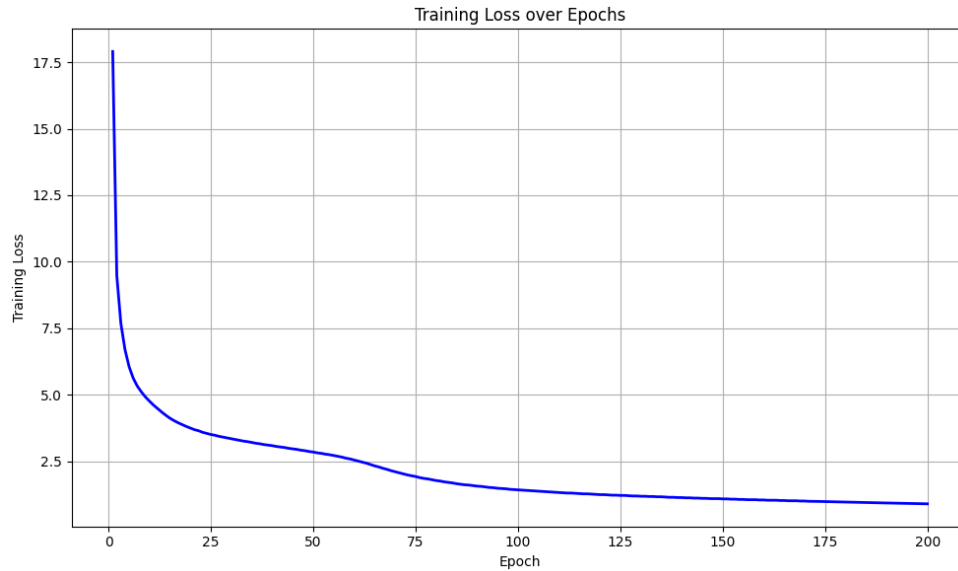
- **Dataset and preprocessing:** The model was trained and evaluated using 7-second music excerpts provided by the musdb package. A data pipeline was implemented within the dataio prep function to create manifest dictionaries for the training, validation, and test sets. This pipeline included a function to load the mixture, vocals, and drums audio signals.
- **Model Architecture:** The core of the model was a Dual-Path network consisting of an Encoder that processed the input mixture using a kernel size of 16 and outputting 64 channels with a stride of 8. This

was followed by single-layer Intra-Chunk (SBtfintra) and Inter-Chunk (SBtfinder) Transformer Blocks with a model dimension of 64, a single attention head, and a feed-forward network dimension of 64. Positional encoding and layer normalization were applied within these blocks. The encoded features were then passed to the MaskNet, a Dual Path Model configured to separate two sources (num-source-sep: 2 for vocals and drums in this initial setup), processing 64 input channels to 64 output channels with an intermediate segment length (K) of 250. Finally, a Decoder with an input of 64 channels, an output of 1 channel per source, a kernel size of 16, and a stride of 8 reconstructed the separated audio signals.

- **Training and Testing:** The model was trained and validated for 200 epochs and subsequently evaluated on the test set. The Signal-to-Interference-plus-Noise Ratio (SI-SNR) was used as the primary loss function during training. The performance on the test set was assessed using metrics such as SI-SNR and Signal-to-Distortion Ratio (SDR).

1.1.3 Results

- **Mean SI-SNR:** -0.25 dB
- **Mean SI-SNR Improvement (SI-SNRi):** 4.51 dB
- **Mean SDR:** 0.06 dB
- **Mean SDR Improvement (SDRi):** 5.14 dB



1.1.4 Analysis

The baseline model, trained for 200 epochs on the 7-second excerpts from the MUSDB18 dataset, demonstrated a modest degree of source separation. The positive Mean SI-SNR Improvement (4.51 dB) and Mean SDR Improvement (5.14 dB) indicate that the model learned to reduce interference and distortion to some extent compared to the original mixture. However, the absolute Mean SI-SNR (-0.25 dB) and Mean SDR (0.06 dB) values suggest that the quality of the separated vocals and drums remains limited, with significant room for improvement.

The consistent decrease in training loss over the 200 epochs, reaching a final value of approximately 0.90, suggests that the model effectively learned the characteristics of the training data. This ability to overfit the small, 7-second excerpts serves as a basic validation of the training pipeline's functionality.

Despite the learning observed on the training set, the low absolute SI-SNR and SDR scores on the test set indicate that the model's ability to generalize to unseen short excerpts is limited. This suggests that the model architecture or the limited training data might be insufficient to capture the complex relationships required for high-quality source separation, even on short segments.

1.1.5 Limitations

The baseline model's performance was subject to several key limitations. Firstly, the training was conducted on a small subset of the MUSDB18 dataset, which inherently limits the model's ability to generalize to the full complexity of musical recordings. Secondly, the chosen compute platform, Paperspace, imposed restrictions on FFMPEG usage, complicating the efficient handling of the original .stems format. This necessitated alternative data loading approaches. Thirdly, utilizing the musdb package for on-the-fly stem extraction introduced a substantial CPU overhead during the data loading process, even with parallel workers, which further constrained the amount of data that could be effectively processed within the available time. Finally, limited GPU VRAM required a significant reduction in the model's architectural complexity, including decreasing the number of layers in the Transformer blocks and the dimensionality of the internal feature representations. These architectural constraints likely restricted the model's capacity to learn intricate audio patterns and achieve higher separation quality.

1.2 Experiment 2: Impact of Model Dimensionality on Vocal Isolation

1.2.1 Motivation

Building upon the successful establishment of a functional training pipeline in the first experiment using a small subset of the MUSDB18 dataset, the goal of this second experiment was to investigate the influence of key model parameters and the number of training epochs on vocal isolation performance using a larger, custom-prepared MUSDB18 dataset. Specifically, this experiment aimed to explore how varying the dimensionality of the encoder output (`N_encoder_out`), the model’s internal channels (`out_channels`), and the feed-forward network dimension within the Transformer blocks (`d_ffn`), as well as the duration of training, affects the model’s ability to separate vocal tracks from the accompanying instrumentation. By training models with different configurations of these parameters and for varying numbers of epochs on a more substantial portion of the dataset, we sought to gain insights into the model’s capacity and identify potentially more effective parameter settings and training schedules for improved vocal isolation.

1.2.2 Methodology

The methodology for the second experiment largely mirrored that of the first, with the crucial distinction being the utilization of a custom-prepared dataset. This dataset comprised individual `.wav` audio files for each source within sample music tracks. The training set consisted of 50 music audio samples, while the testing set included 18 samples.

- **Dataset and Preprocessing:** The model was trained on randomly extracted chunks of audio from the training samples. The training sequence length supported by the GPU was set to `training_signal_len: 1600000` at a sampling rate of 44.1 kHz. This corresponds to a chunk duration of:

$$\frac{1600000}{44100} \approx 36.28 \text{ seconds}$$

The data pipeline, implemented within the `dataio_prep` function, was adapted to load these `.wav` files for the mixture and individual sources. The model separates 2 sources (vocals and drums) from the mixture sample.

- **Model Architecture:** We initially employed a Dual-Path network architecture with reduced dimensionality to accommodate computational constraints. The initial configuration was:

- `N_encoder_out: 32`
- `out_channels: 32`

- `d_ffn`: 32

The number of data loader workers was initially set to 3 to potentially accelerate data loading and reduce training time. However, for a subsequent training run, the number of workers was reduced to 1, and the model dimensionality was increased to:

- `N_encoder_out`: 64
- `out_channels`: 64
- `d_ffn`: 64

This adjustment aimed to explore the impact of increased model capacity on the separation performance. The remaining architectural components, including the Encoder (kernel size 16, stride 8), single-layer Intra-Chunk and Inter-Chunk Transformer Blocks (with `d_model` equal to `out_channels` and 1 attention head), the MaskNet (configured for two sources), and the Decoder (kernel size 16, stride 8), remained consistent with the baseline experiment.

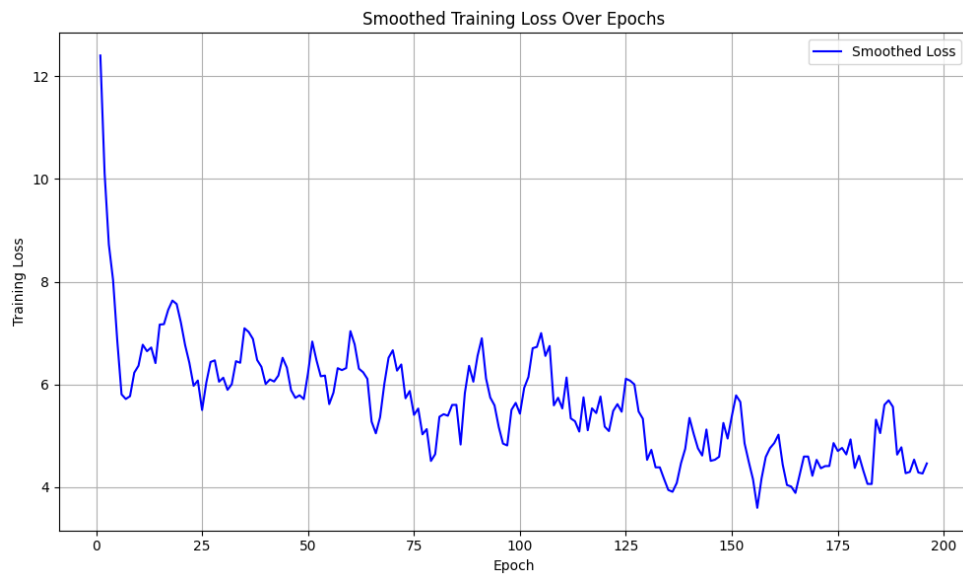
- **Training and Testing:** The first model architecture (with lower dimensionality) was trained for both 200 and 500 epochs. The second model architecture (with higher dimensionality) was trained for 200 epochs. Evaluation on the 18-sample test set was performed using the Mean Signal-to-Distortion Ratio (SDR), Mean Signal-to-Artifacts Ratio (SAR), and Mean Signal-to-Interference Ratio (SIR) as the primary evaluation metrics.

1.2.3 Results

This experiment explored the impact of model dimensionality and training duration on vocal isolation performance. The results for the different configurations are as follows:

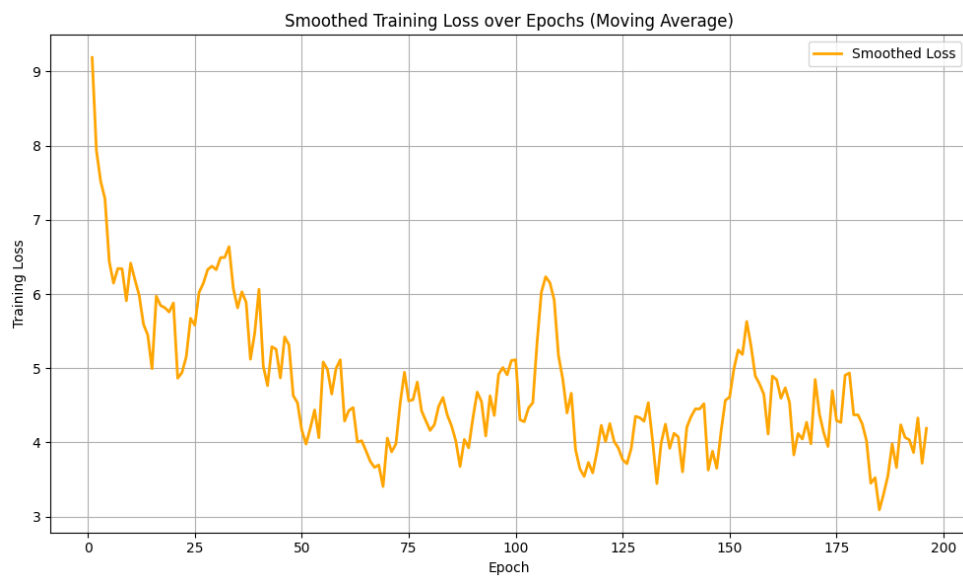
Lower Dimensionality Model (200 Epochs)

- **Mean SDR:** -0.95 dB
- **Mean SIR:** 11.60 dB
- **Mean SAR:** -0.12 dB

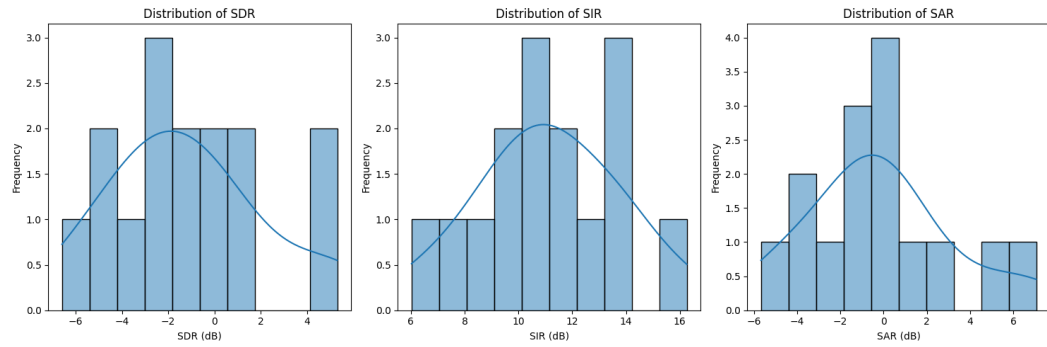


Higher Dimensionality Model (200 Epochs)

- Mean SDR: -1.16 dB
- Mean SIR: 11.13 dB
- Mean SAR: -0.17 dB



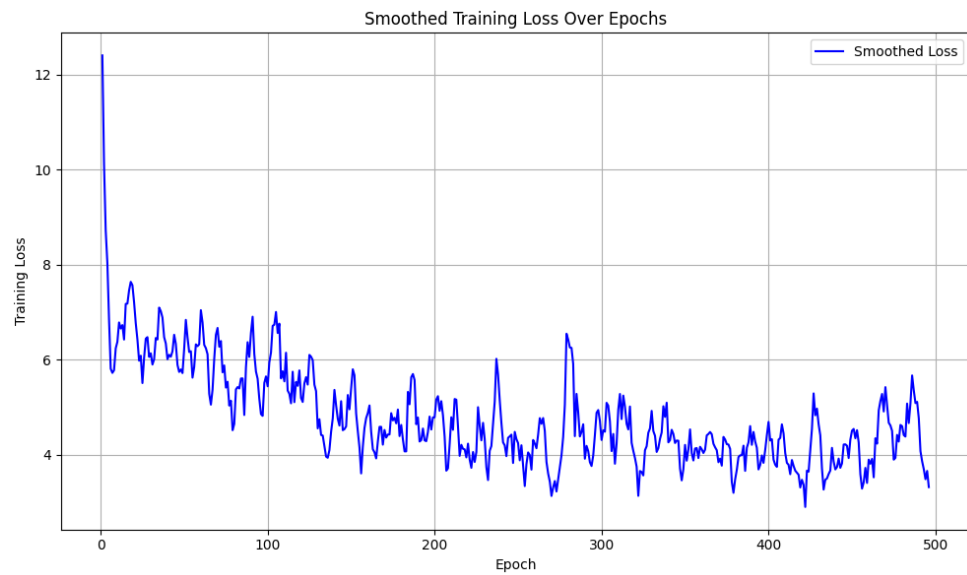
Training Loss for Higher Dimensionality Model (200 Epochs)

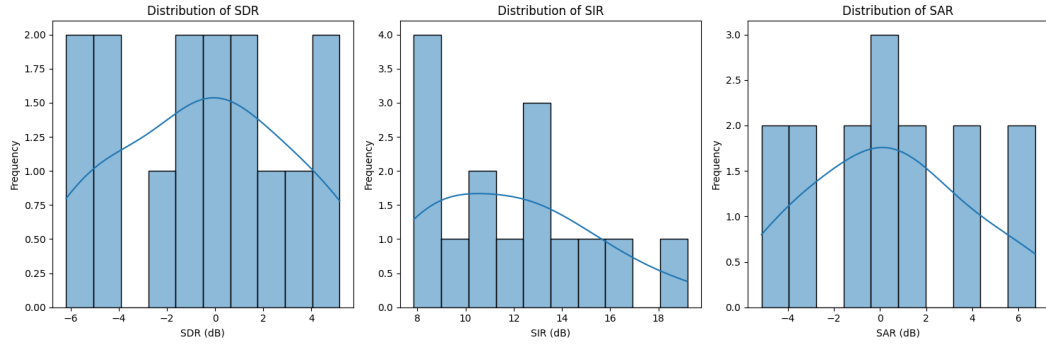


Distribution of SDR, SIR SAR metrics

Lower Dimensionality Model (500 Epochs)

- **Mean SDR:** -0.46 dB
- **Mean SIR:** 11.99 dB
- **Mean SAR:** 0.37 dB





Distribution of SDR, SIR SAR metrics

Additional insights, including per-song breakdowns of SDR, SAR, and SIR metrics for two-stem models, is provided in Appendix Figures 2 and 3 respectively.

1.2.4 Analysis

This experiment investigated the impact of model dimensionality and training duration on vocal isolation performance using a custom dataset of full-length music tracks. The results indicate a complex relationship between these factors and the achieved separation quality.

For the lower dimensionality model trained for 200 epochs, the Mean SDR was -0.95 dB, Mean SIR was 11.60 dB, and Mean SAR was -0.12 dB. Extending the training of this model to 500 epochs resulted in improvements across all metrics, with the Mean SDR increasing to -0.46 dB, Mean SIR to 11.99 dB, and Mean SAR to 0.37 dB. This suggests that longer training can benefit even lower-capacity models.

Interestingly, the higher dimensionality model trained for 200 epochs yielded slightly worse performance than the lower dimensionality model trained for 500 epochs, with a Mean SDR of -1.16 dB, Mean SIR of 11.13 dB, and Mean SAR of -0.17 dB. This indicates that simply increasing model capacity without sufficient training or optimized hyperparameters may not lead to immediate improvements and could even hinder performance.

The provided training loss values for the higher dimensionality model over 200 epochs exhibit a general downward trend, suggesting learning. However, the loss trajectory appears unstable and jittery, with noticeable fluctuations throughout the training. This erratic behavior is likely attributable to the training strategy of using random chunks from the full-length audio samples in each epoch. As the model encounters different, potentially unseen, segments of the music in later epochs, the loss can temporarily increase as it adapts to new acoustic characteristics. This phenomenon highlights the variability inherent in training on randomly sampled chunks from longer audio signals.

The overall trend of decreasing loss with more epochs (observed in the

lower dimensionality model going from 200 to 500 epochs) suggests that continued training allows the model to better generalize across the diverse acoustic landscapes present in the training data, despite the per-epoch variability introduced by random chunking. The fact that the lower dimensionality model trained for longer outperformed the higher dimensionality model trained for fewer epochs underscores the importance of both model capacity and sufficient training duration for effective music source separation.

1.2.5 Limitations

Several limitations affected this experiment. Firstly, despite having more computational resources than the baseline, GPU VRAM constraints still necessitated a reduction in the model’s intended architectural complexity. Secondly, to keep training times manageable, we employed random chunking of audio within each epoch, rather than processing all possible segments. This approach might have introduced variability in the training process, as the model didn’t see all parts of each song in every epoch. Thirdly, we observed instances where some randomly selected audio chunks contained all-zero reference sources. This issue violates the assumption of non-silent ground truth during evaluation and could have introduced inaccuracies or ambiguities in the calculated SDR, SIR, and SAR metrics. Addressing these limitations, such as using more sophisticated training strategies for long audio, and ensuring clean, non-silent reference data, could lead to more robust and reliable results in future experiments.

1.3 Experiment 3: Impact of Separation Complexity (Number of Sources) on Vocal Isolation

1.3.1 Motivation

The primary motivation for this third experiment was to investigate the impact of the separation task’s complexity on the model’s ability to isolate vocals. Specifically, we aimed to compare the performance of a model trained to separate only two sources – vocals and drums – against a model trained to separate all four standard stems present in the MUSDB18 dataset: vocals, drums, bass, and other. By training and evaluating these two models on the same dataset and using a consistent underlying architecture, we sought to determine whether simplifying the separation target to a smaller number of sources would lead to improved performance specifically on the vocals, or if a model trained on the more complex four-stem separation task would learn more robust and generalizable features beneficial for isolating individual sources as well. This comparison provides insights into how the model’s learning is affected by the number of target sources and the potential trade-offs between task simplicity and overall source separation capability.

1.3.2 Methodology

The methodology for this third experiment shared several similarities with the second experiment, particularly in the underlying model architecture and evaluation metrics.

- **Dataset and Preprocessing:** Similar to Experiment 2, the model was trained and evaluated using a custom-prepared dataset of individual `.wav` audio files for each source within sample music tracks. The training was performed on randomly extracted chunks of audio with a training sequence length of 1600000 samples at a 44.1 kHz sampling rate (approximately 36.28 seconds). However, in contrast to Experiment 2, the data pipeline within the `dataio_prep` function was configured to load the mixture and all four individual stems: vocals, drums, bass, and other. The model was therefore trained to separate these four sources from the mixed audio. The testing procedure also involved processing mixtures from the test set and evaluating the separation performance for each of the four stems.
- **Model Architecture:** The model architecture employed in this experiment was the same higher-dimensionality Dual-Path network configuration used in the second part of Experiment 2:
 - `N_encoder_out`: 64
 - `out_channels`: 64
 - `d_ffn`: 64

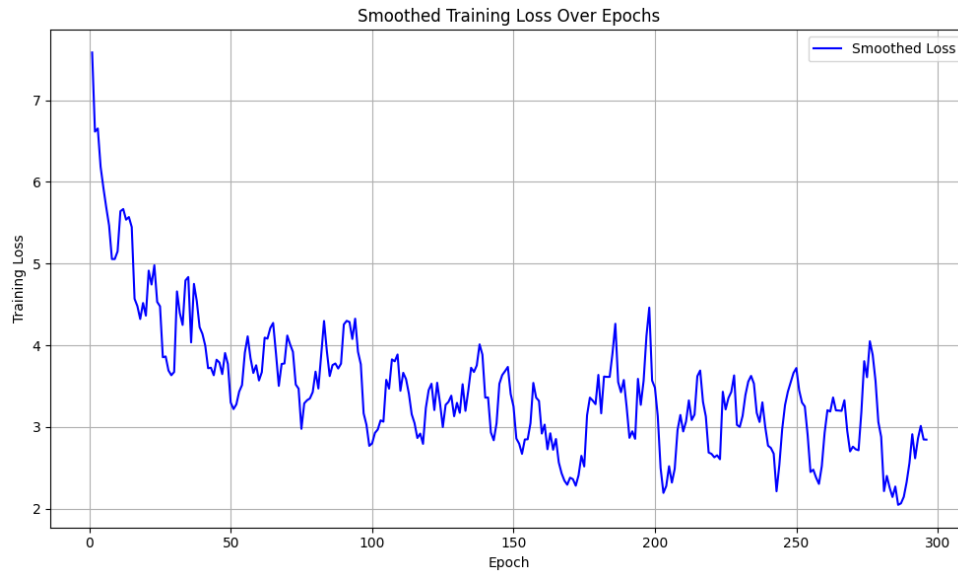
The remaining architectural components (Encoder, Transformer Blocks, MaskNet, Decoder) were also identical to those described in the Model Architecture section of Experiment 2. The key difference in this experiment’s model configuration was the MaskNet being configured to separate four sources (`num_source_seperation:4`).

- **Training and Testing:** The model in this experiment was trained and validated for 300 epochs. Evaluation on the same 18-sample test set used in Experiment 2 was performed using the Mean Signal-to-Distortion Ratio (SDR), Mean Signal-to-Artifacts Ratio (SAR), and Mean Signal-to-Interference Ratio (SIR) as the primary evaluation metrics, calculated for each of the four separated stems (vocals, drums, bass, other).

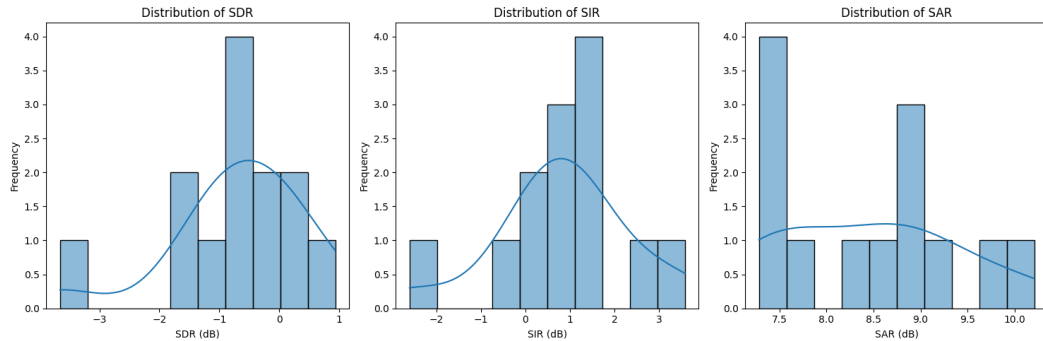
1.3.3 Results

The model trained to separate all four stems (vocals, drums, bass, and other) for 300 epochs achieved the following average performance metrics on the test set:

- **Mean SDR:** -0.70 dB
- **Mean SIR:** 0.82 dB
- **Mean SAR:** 8.45 dB



Training Loss for Four-Stem Separation Model (300 Epochs)



Distribution of SDR, SIR SAR metrics

Additional insights, including per-song breakdowns of SDR, SAR, and SIR metrics for four-stem model, is provided in Appendix Figure 1

1.3.4 Analysis

This experiment compared a model trained to separate four stems (vocals, drums, bass, other) against the two-stem (vocals, drums) separation of the higher-dimensional model in Experiment 2. The four-stem model was trained for 300 epochs, while the two-stem model was trained for 200 epochs.

The four-stem model achieved average metrics of SDR -0.70 dB, SIR 0.82 dB, and SAR 8.45 dB across all stems. Comparing these averages to the two-stem model (SDR -1.16 dB, SIR 11.13 dB, SAR -0.17 dB), the four-stem model shows a higher average SDR and SAR, but a lower average SIR.

Considering the longer training duration of the four-stem model, the improved average SDR and SAR might be attributed to the additional training epochs allowing for better convergence. However, the lower average SIR suggests that separating more sources simultaneously could lead to increased interference between the stems, even with more training. The distribution of the model's capacity across four outputs, as opposed to two, likely also plays a significant role in the observed differences.

2 Conclusion and Future Work

2.1 Summary of your work and your findings

This project focused on developing and evaluating deep learning models for vocal isolation in music, employing a Dual-Path network architecture inspired by SpeechBrain. We conducted three primary experiments. The first established a baseline using short MUSDB18 excerpts, confirming pipeline functionality but revealing limited separation quality due to data and computational constraints. The second experiment investigated the impact of model dimensionality and training duration on a custom dataset of longer audio, finding that extended training generally improved performance, while increased dimensionality alone was insufficient. The third experiment explored the effect of separation complexity by comparing two-stem (vocals, drums) versus four-stem (vocals, drums, bass, other) separation, suggesting a trade-off in performance characteristics. Overall, this work demonstrates the feasibility of deep learning for vocal isolation while underscoring the importance of dataset characteristics, computational resources, model architecture, training strategies, and the complexity of the separation task in achieving effective results.

2.2 Future Work

Several avenues exist for future work to build upon the findings of this project and address the identified limitations. To enhance the model's performance, increasing its capacity by exploring deeper architectures with more layers in the Transformer blocks and the encoder, as well as increasing the number of attention heads, could be beneficial. Utilizing the full MUSDB18 dataset for training, and potentially employing strategies to process longer audio sequences more effectively (rather than relying solely on random chunking), could lead to improved generalization and separation quality across entire songs.

Furthermore, future research could investigate more sophisticated training techniques, such as curriculum learning or different loss functions tailored for audio separation. Addressing the issue of all-zero reference sources during evaluation is also crucial for obtaining more reliable and interpretable metrics. Finally, exploring alternative model architectures beyond the Dual-Path network, such as U-Nets or other state-of-the-art source separation models, and conducting a more comprehensive hyperparameter search, could potentially yield significant improvements in vocal isolation performance. Qualitative evaluations through listening tests should also be incorporated to provide a more subjective assessment of the separation quality.

References

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- [3] RAFII, Z., LIUTKUS, A., STÖTER, F.-R., MIMILAKIS, S. I., AND BITTNER, R. The MUSDB18 corpus for music separation.
- [4] RAVANELLI, M., PARCOLLET, T., PLANTINGA, P., ROUHE, A., CORNELL, S., DAWALATABAD, N., HEBA, A., ZHONG, J., CHANG, J.-C., YEH, S.-L., ET AL. SpeechBrain: Open-Source Conversational AI for Everyone. *arXiv preprint arXiv:2109.01501* (2021).

3 Appendix

3.1 Per-song Separation Metrics

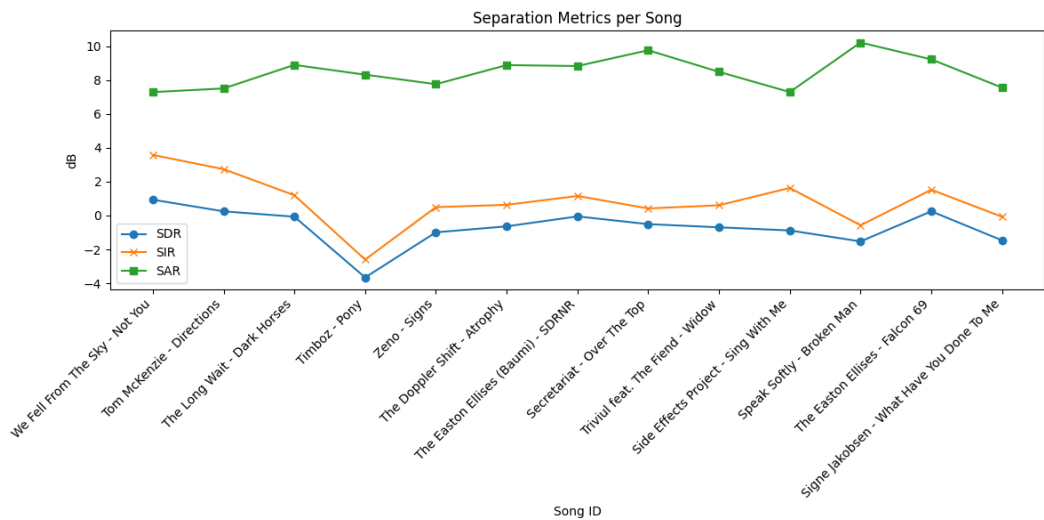


Figure 1: Separation metrics by song for four-stem model (300 epochs).

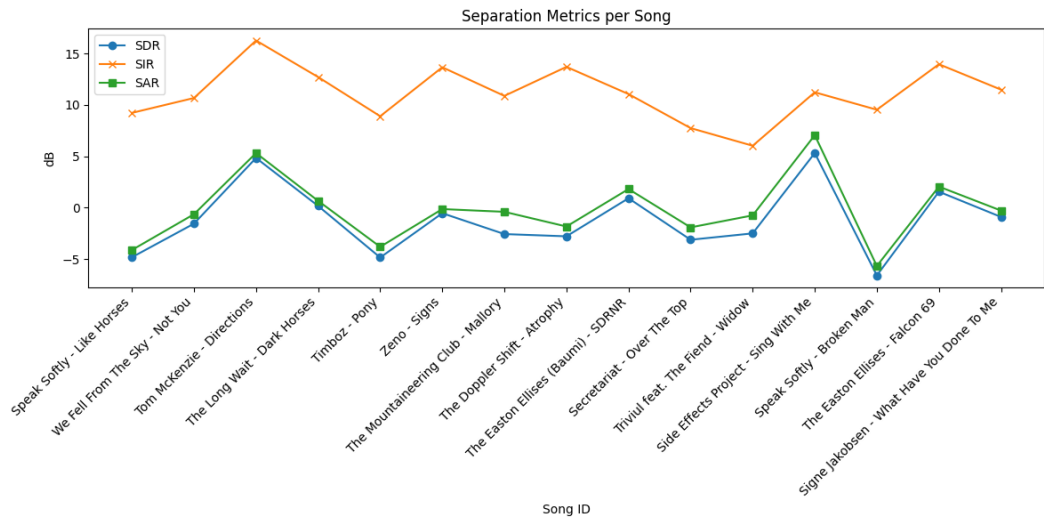


Figure 2: Separation metrics by song for higher-dimensional model (200 epochs).

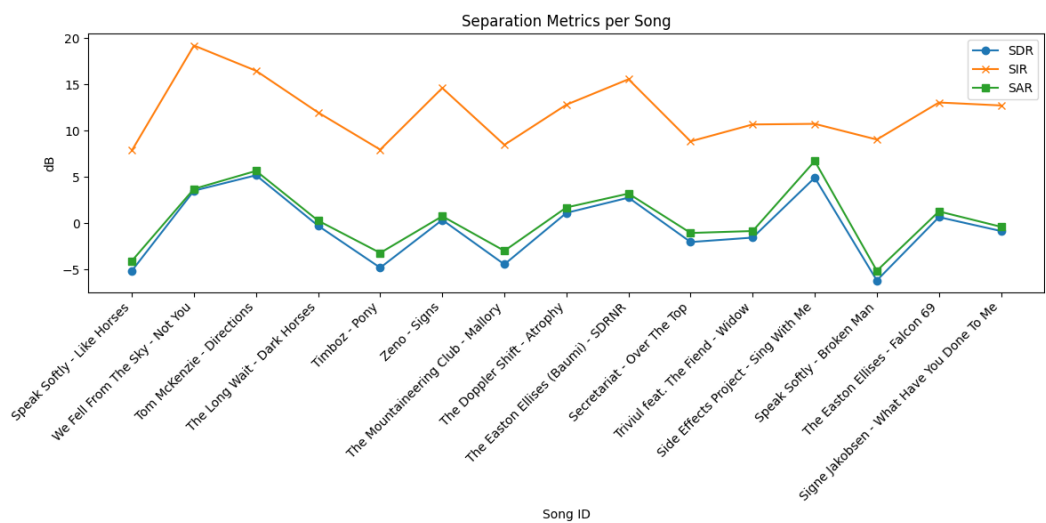


Figure 3: Separation metrics by song for lower-dimensional model (500 epochs).